Sherry Yang

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Google Scholar

Employment

September 2025 **Assistant Professor**, New York University, Computer Science (Courant)
Research in reinforcement learning, world models, agents, and robotics.

2022-Now **Staff Research Scientist**, Google DeepMind, Manager: Dale Schuurmans
Researching in reinforcement learning and foundation models for decision making.

2018-2022 **Research Scientist**, Google Brain,
Researched in reinforcement learning, representation learning, generative modeling.

Education

2024-2025 **Post-doc in Computer Science**, Stanford University, 🎓Percy Liang
Stanford NLP Group. RL for machine learning engineering agents.

2021-2024 **Ph.D. in Computer Science**, UC Berkeley, 🎓Pieter Abbeel
Berkeley Artificial Intelligence Research (BAIR).

2017-2018 **M.Eng. in Computer Science**, MIT, 🎓Julian Shun
Computer Science and Artificial Intelligence Laboratory (CSAIL).

2014-2017 **B.S. in Computer Science**, MIT, 🎓Patrick Winston
Also worked with Daniel Jackson and Saman Amarasinghe.

Publications and Preprints

(* indicates equal contribution)

Preprint26 Yuyang Wu, Stefano Falletta, Delia McGrath, **Sherry Yang**. *CrystalReasoner: Reasoning and RL for Property-Conditioned Crystal Structure Generation.* Preprint, 2026.

Preprint26 Joy He-Yueya, Anikait Singh, Ge Gao, Michael Y Li, **Sherry Yang**, Chelsea Finn, Emma Brunskill, Noah D Goodman. *GIANTS: Generative Insight Anticipation from Scientific Literature.* Preprint, 2026.

Preprint26 Omar Shaikh, Valentin Teutschbein, Kanishk Gandhi, Yikun Chi, Nick Haber, Thomas Robinson, Nilam Ram, Byron Reeves, **Sherry Yang**, Michael S Bernstein, Diyi Yang. *Learning Next Action Predictors from Human-Computer Interaction.* Preprint, 2026.

Preprint26 Ansh Kumar Sharma, Yixiang Sun, Ninghao Lu, Yunzhe Zhang, Jiarao Liu, **Sherry Yang**. *World-Gymnast: Training Robots with Reinforcement Learning in a World Model.* Preprint, 2026.

Preprint26 **Sherry Yang**. *World Models as an Intermediary between Agents and the Real World.* Preprint, 2026.

Preprint25 Gemini Robotics Team, Krzysztof Choromanski, Coline Devin, Yilun Du, Debidatta Dwibedi, Ruiqi Gao, Abhishek Jindal, Thomas Kipf, Sean Kirmani, Isabel Leal, Fangchen Liu, Anirudha Majumdar, Andrew Marmon, Carolina Parada, Yulia Rubanova, Dhruv Shah, Vikas Sindhwani, Jie Tan, Fei Xia, Ted Xiao, **Sherry Yang**, Wenhao Yu, Allan Zhou. *Evaluating Gemini Robotics Policies in a Veo World Simulator.* Preprint, 2025.

ICLR26 Rushi Qiang, Yuchen Zhuang, Anikait Singh, Percy Liang, Chao Zhang, **Sherry Yang**, Bo Dai. *Mle-smith: Scaling MLE Tasks with Automated Multi-Agent Pipeline.* International Conference on Learning Representations (ICLR), 2026.

Preprint25 Gemini Robotics Team. *Gemini Robotics 1.5: Pushing the Frontier of Generalist Robots with Advanced Embodied Reasoning, Thinking, and Motion Transfer.* Preprint, 2025.

ICLR26 **Sherry Yang**, Joy He-Yueya, Percy Liang. *Reinforcement Learning for Machine Learning Engineering Agents.* International Conference on Learning Representations (ICLR), 2026.

Preprint25 Gemini Team. *Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities.* Preprint, 2025.

NeurIPS25 Zhao-Heng Yin, **Sherry Yang**, Pieter Abbeel. *Object-Centric 3D Motion Field for Robot Learning from Human Videos.* Advances in Neural Information Processing Systems (NeurIPS), 2025.

ICLR26 Julian Quevedo, Ansh Kumar Sharma, Yixiang Sun, Percy Suryavanshi, Varad Liang, **Sherry Yang**. *WorldGym: World Model as An Environment for Policy Evaluation.* International Conference on Learning Representations (ICLR), 2026.

ICLR26 Rushi Qiang, Yuchen Zhuang, Yinghao Li, Rongzhi Zhang, Changhao Li, Ian Shu-Hei Wong, **Sherry Yang**, Percy Liang, Chao Zhang, Bo Dai. *Mle-dojo: Interactive Environments for Empowering LLM Agents in Machine Learning Engineering.* International Conference on Learning Representations (ICLR), 2026.

NeurIPS25 Rushi Qiang, Yuchen Zhuang, Yinghao Li, Rongzhi Zhang, Changhao Li, Ian Shu-Hei Wong, **Sherry Yang**, Percy Liang, Chao Zhang, Bo Dai. *Mle-dojo: Interactive Environments for Empowering LLM Agents in Machine Learning Engineering.* Advances in Neural Information Processing Systems (NeurIPS), 2025.

NatCompSci25 Theo Jaffrelot Inizan, **Sherry Yang**, Aaron Kaplan, Yen-hsu Lin, Jian Yin, Saber Mirzaei, Mona Abdelgaid, Ali H Alawadhi, KwangHwan Cho, Zhiling Zheng, Ekin Dogus Cubuk, Christian Borgs, Jennifer T Chayes, Kristin A Persson, Omar M Yaghi. *System of Agentic AI for the Discovery of Metal-Organic Frameworks.* Nature Computational Science, 2025.

Preprint25 Theo Jaffrelot Inizan, Aaron Kaplan, **Sherry Yang**, Yen-hsu Lin, Mona Abdelgaid, Jian Yin, Zhiling Zheng, Saber Mirzaei, Ali H Alawadhi, Ekin D Cubuk, Christian Borgs, Jennifer T Chayes, Kristin A Persson, Omar M Yaghi. *Generative Model for Enhancing Reticular Material Discovery.* Preprint, 2025.

NeurIPS24 Hanjun Dai, Bethany Y Wang, Xingchen Wan, Bo Dai, **Sherry Yang**, Azade Nova, Pengcheng Yin, Phitchaya M Phothilimthana, Charles Sutton, Dale Schuurmans. *UQE: A Query Engine for Unstructured Databases.* Advances in Neural Information Processing Systems (NeurIPS), 2024.

JRML24 Hiroki Furuta, Heiga Zen, Dale Schuurmans, Aleksandra Faust, Yutaka Matsuo, Percy Liang, **Sherry Yang**. *Improving Dynamic Object Interactions in Text-to-Video Generation with AI Feedback.* Journal of Reinforcement and Machine Learning, 2024.

Preprint24 Achint Soni, Sreyas Venkataraman, Abhranil Chandra, Sebastian Fischmeister, Percy Liang, Bo Dai, **Sherry Yang**. *VideoAgent: Self-Improving Video Generation.* Preprint, 2024.

NeurIPS24 **Sherry Yang**, Simon Batzner, Ruiqi Gao, Muratahan Aykol, Alexander L Gaunt, Brendan McMorrow, Danilo J Rezende, Dale Schuurmans, Igor Mordatch, Ekin D Cubuk. *Generative Hierarchical Materials Search.* Advances in Neural Information Processing Systems (NeurIPS), 2024.

NeurIPS24 Shicong Cen, Jincheng Mei, Katayoon Goshvadi, Hanjun Dai, Tong Yang, **Sherry Yang**, Dale Schuurmans, Yuejie Chi, Bo Dai. *Value-Incentivized Preference Optimization: A Unified Approach to Online and Offline RLHF.* Advances in Neural Information Processing Systems (NeurIPS), 2024.

Preprint24 Gemini Team. *Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context.* Preprint, 2024.

Preprint **Sherry Yang**, Jacob Walker, Jack Parker-Holder, Yilun Du, Jake Bruce, Andre Barreto, Pieter Abbeel, Dale Schuurmans. *Video as the New Language for Real-World Decision Making.*

Preprint David Venuto, Sami Nur Islam, Martin Klissarov, Doina Precup, **Sherry Yang**, Ankit Anand. *Code as Reward: Empowering Reinforcement Learning with VLMs.*

ICLR24 **Sherry Yang**, Yilun Du, Kamyar Ghasemipour, Jonathan Tompson, Dale Schuurmans, Pieter Abbeel. *Learning Interactive Real-World Simulators.* International Conference on Learning Representations (ICLR), 2024, **Oral**. NeurIPS workshop on Instruction Tuning and Instruction Following, 2023. **Best Paper**.

ICLR24 **Sherry Yang**, KwangHwan Cho, Amil Merchant, Pieter Abbeel, Dale Schuurmans, Igor Mordatch, Ekin Dogus Cubuk. *Scalable Diffusion for Materials Generation.* International Conference on Learning Representations (ICLR), 2024. AI for Accelerated Materials Design Workshop at NeurIPS 2023, **Spotlight**.

ICLR24 **Sherry Yang***, Yilun Du*, Bo Dai, Dale Schuurmans, Joshua B Tenenbaum, Pieter Abbeel. *Probabilistic Adaptation of Text-to-Video Models.* International Conference on Learning Representations (ICLR), 2024.

Preprint23 **Sherry Yang**, Ofir Nachum, Yilun Du, Jason Wei, Pieter Abbeel, Dale Schuurmans. *Foundation Models for Decision Making: Problems, Methods, and Opportunities.* Preprint.

ICLR24 Yilun Du, **Sherry Yang**, Pete Florence, Fei Xia, Ayzaan Wahid, Brian Ichter, Pierre Sermanet, Tianhe Yu, Pieter Abbeel, Joshua B Tenenbaum, Leslie Kaelbling, Andy Zeng, Jonathan Tompson. *Video Language Planning. Learning Interactive Real-World Simulators.* International Conference on Learning Representations (ICLR), 2024.

NeurIPS23 Yilun Du*, **Sherry Yang***, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans, Pieter Abbeel. *Learning Universal Policies via Text-Guided Video Generation.* Advances in Neural Information Processing Systems (NeurIPS), 2023. **Spotlight**.

ICML23 David Venuto*, **Sherry Yang***, Pieter Abbeel, Doina Precup, Igor Mordatch, Ofir Nachum. *Multi-Environment Pretraining Enables Transfer to Action Limited Datasets.* International Conference on Machine Learning (ICML), 2023.

ICLR23 **Sherry Yang**, Dale Schuurmans, Pieter Abbeel, Ofir Nachum. *Dichotomy of control: Separating what you can control from what you cannot.* International Conference on Learning Representations (ICLR), 2023, **Oral**.

ICLR23 Charlie Snell, Ilya Kostrikov, Yi Su, **Sherry Yang**, Sergey Levine. *Offline RL for Natural Language Generation with Implicit Language Q Learning.* International Conference on Learning Representations (ICLR), 2023.

NeurIPS22 **Sherry Yang**, Dale Schuurmans, Pieter Abbeel, Ofir Nachum. *Chain of Thought Imitation Learning with Procedure Cloning.* Advances in Neural Information Processing Systems (NeurIPS), 2022.

NeurIPS22 Kuang-Huei Lee, Ofir Nachum, **Sherry Yang**, Lisa Lee, Daniel Freeman, Winnie Xu, Sergio Guadarrama, Ian Fischer, Eric Jang, Henryk Michalewski, Igor Mordatch. *Multi-Game Decision Transformers.* Advances in Neural Information Processing Systems (NeurIPS), 2022, **Spotlight**.

ICML22 Tianjun Zhang, Tongzheng Ren, **Sherry Yang**, Joseph Gonzalez, Dale Schuurmans, Bo Dai. *Making linear mdps practical via contrastive representation learning.* International Conference on Machine Learning (ICML), 2022.

ICML22 Hanjun Dai, **Sherry Yang**, Yuan Xue, Dale Schuurmans, Bo Dai. *Marginal Distribution Adaptation for Discrete Sets via Module-Oriented Divergence Minimization.* International Conference on Machine Learning (ICML), 2022.

AISTATS22 **Sherry Yang***, Bo Dai*, Ofir Nachum*, George Tucker, Dale Schuurmans. *Offline Policy Selection under Uncertainty.* Preprint:2012.06919, 2022.

ICLR22 **Sherry Yang**, Sergey Levine, Ofir Nachum. *Trail: Near-Optimal Imitation Learning with Suboptimal Data.* International Conference on Learning Representations (ICLR), 2022.

NAACL21 Charlie Snell, **Sherry Yang**, Justin Fu, Yi Su, Sergey Levine. *Context-Aware Language Modeling for Goal-Oriented Dialogue Systems .* North American Chapter of the ACL (NAACL), 2022

NAACL21	Siddharth Verma, Justin Fu, **Sherry Yang**, Sergey Levine. *CHAI: A CHatbot AI for Task-oriented Dialogue with Offline Reinforcement Learning.* North American Chapter of the ACL (NAACL), 2022, **Oral**.
NeurIPS21	Hongyu Ren, Hanjun Dai, Zihang Dai, **Sherry Yang**, Jure Leskovec, Dale Schuurmans, Bo Dai. *Combiner: Full Attention Transformer with Sparse Computation Cost.* Conference on Neural Information Processing Systems (NeurIPS), 2021
NeurIPS21	Ofir Nachum, **Sherry Yang**. *Provable Representation Learning for Imitation with Contrastive Fourier Features.* Conference on Neural Information Processing Systems (NeurIPS), 2021
EMNLP21	Haoming Jiang, Bo Dai, **Sherry Yang**, Wei Wei, Tuo Zhao. *Towards Automatic Evaluation of Dialog Systems: A Model-Free Off-Policy Evaluation Approach .* The Conference on Empirical Methods in Natural Language Processing (EMNLP), 2021
ICML21	**Sherry Yang**, Ofir Nachum. *Representation Matters: Offline Pretraining for Sequential Decision Making.* International Conference on Machine Learning (ICML), 2021.
ICLR21	Justin Fu, Mohammad Norouzi, Ofir Nachum, George Tucker, Ziyu Wang, Alexander Novikov, **Sherry Yang**, R. Michael Zhang, Yutian Chen, Aviral Kumar, Cosmin Paduraru, Sergey Levine, Thomas Paine. *Benchmarks for Deep Off-Policy Evaluation.* International Conference on Learning Representations (ICLR), 2021.
NeurIPS20	**Sherry Yang***, Ofir Nachum*, Bo Dai*, Lihong Li, Dale Schuurmans. *Off-Policy Evaluation via the Regularized Lagrangian.* Conference on Neural Information Processing Systems (NeurIPS), 2020.
ICML20	**Sherry Yang***, Bo Dai*, Hanjun Dai, Dale Schuurmans. *Energy-Based Processes for Exchangeable Data.* International Conference on Machine Learning (ICML), 2020.
HCML19	**Sherry Yang**, Been Kim. *Benchmarking Attribution Methods with Relative Feature Importance.* NeurIPS workshop on Human-Centric Machine Learning, 2019, **Oral**.
OOPSLA18	Yunming Zhang, **Sherry Yang**, Riyadh Baghdadi, Shoaib Kamil, Julian Shun, Saman Amarasinghe. *GraphIt: A High-Performance Graph DSL.* The Object-Oriented Programming, Systems, Languages and Applications (OOPSLA), 2018.

Patents

Filed 2023	Ruben Cho, Ekin Dogus Cubuk, Igor Mordatch, Dale Schuurmans, Amil Merchant, Sherry Yang. *Predicting Structures of Substances Using Diffusion Models.* US63/541,255. Filed 2023-09-28.
Filed 2023	Sherry Yang, Yilun Du, Dale Schuurmans, Bo Dai. *Generating Domain-Specific Videos Using Diffusion Models.* US63/502,885. Filed 2023-05-17.
Filed 2023	Hanjun Dai, Sherry Yang, Emily Xue, Dale Schuurmans, Bo Dai. *Generative Models for Discrete Datasets Constrained by a Marginal Distribution Specification.* US17/951,889. Filed 2022-09-23.
2023	Hanjun Dai, Zihang Dai, Sherry Yang, Hongyu Ren, Dale Schuurmans, Bo Dai. 2023. *Full Attention with Sparse Computation Cost.* US17/860,691. Filed 2022-07-08. 2023-01-26.
2022	Sherry Yang, Hanjun Dai, Dale Schuurmans, Bo Dai. 2022. *Energy-Based Processes for Exchangeable Data.* US17/239,320. Filed 2021-04-23. 2022-10-27.

Talks

Apr 2026	ICLR 2026 Workshop on Multimodal Intelligence: Next Token Prediction & Beyond.
Apr 2026	New York University CILVR Seminar.
Apr 2026	Brown University Invited Talk.
Feb 2026	*Learning World Models and Agents for High-Cost Environments.* Harvard Kempner Seminar Series.

Feb 2026	*Pretraining World Models for Post-Training Physical Agents.* World Modeling Workshop, Mila Quebec AI Institute.
Jan 2026	Georgia Tech School of CSE Seminar Series.
Dec 2025	NeurIPS 2025 Workshop on Language models, Agent models, and World models (LAW).
Nov 2025	UCLS NLP Seminar Series.
Nov 2025	Machine Learning at Flatiron Institute Seminar Series.
Oct 2025	Harvard University Kempner Institute Seminar.
Oct 2025	University of Massachusetts Amherst Machine Learning and Friends Lunch.
Oct 2025	Mila Robot Learning Seminar.
Jul 2025	ICML Workshop on Building Physically Plausible World Models.
Jun 2025	CVPR Tutorial on from Video Generation to World Model.
Jun 2025	CVPR Tutorial on Scalable Generative Models in Computer Vision.
Jun 2025	CVPR Workshop on Multimodal Video Agents (LOVE).
Mar 2025	*Harnessing Generative Models for Scalable Materials Discovery.* Generative Modeling for Chemistry, Biology, & Material Discovery Symposium at ACS Spring 2025.
Mar 2025	Berkeley Bakar Institute of Digital Materials for the Planet Seminar.
Nov 2023	*Foundation Models for Decision Making.* Stanford.
Nov 2023	*Foundation Models for Decision Making.* University of Washington.
Oct 2023	*Foundation Models for Decision Making.* UT Austin.
Oct 2023	*Foundation Models for Decision Making.* AI seminar at CMU.
Oct 2023	*Foundation Models for Decision Making.* NYU.
Oct 2023	*Foundation Models for Decision Making.* Princeton.
Oct 2023	*Foundation Models for Decision Making.* CSL seminar at MIT.
Oct 2023	*Foundation Models for Decision Making.* Georgia Tech.
July 2022	*Transformers for RL? Why and How?* Google Brain.
June 2022	*Chain of Thought Imitation with Procedure Cloning.* Google Brain.
March 2021	*Offline Pretraining for Sequential Decision Making.* Google Brain.
June 2020	*Off-Policy Evaluation via the Regularized Lagrangian.* Google Brain.

Awards and Fellowships

2022, 2023	Google-BAIR Commons Research Grant.
2021	Berkeley AI Research (BAIR) Ignition Award.
2017	Grace Hopper Research Scholar Award.
2017	MIT EECS Slaughter Undergraduate Research and Innovation Scholar Award.
2016, 2017	MIT Jack C. Tang Scholar Award.

Current Students

PhD	Tatsuya Kamijo
PhD	Doyoung Kim
PhD	Mattia Scardecchia
PhD	Yunzhe Zhang
Master	Zijin Hu
Master	Ansh Sharma
Master	Yixiang Sun
Undergraduate	Ninghao (Alex) Lu

Undergraduate	Yuyang Wu

Alumni

2023	Zichao Li (PhD at UC Santa Cruz).
2023	KwangHwan Cho (BS/MS at UC Berkeley).
2022	David Venuto (PhD at McGill University).
2021-2022	Charlie Snell (BS at UC Berkeley).
2021	Haoming Jiang (PhD at Georgia Tech).
2021	Siddharth Verma (MS UC Berkeley).

Professional Service

2022, 2023	**Lead Organizer**, Workshop on Foundation Models for Decision Making at NeurIPS 2022 and 2023.
2021	**Program Committee**, International Joint Conferences on Artificial Intelligence (IJCAI).
2019-2023	**Reviewer**, ICLR (2020-2023), ICML (2020-2023), NeurIPS (2020-2022), AISTATS (2021), BayLearn (2021), SIGKDD (2020), AAAI (2019).
2022-2023	**Grant Reviewer**, Google Faculty Research Awards (2022, 2023).

Teaching

Fall 2025	**Special Topics**, Reinforcement Learning with Foundation Models, NYU.
Nov 2023	**Guest Lecturer**, CSCI677 at University of Southern California.
Oct 2023	**Guest Lecturer**, CSE6243 at Georgia Tech.
Fall 2022	**Graduate Student Instructor**, CS 188: Introduction to Artificial Intelligence, UC Berkeley.
Spring 2018	**Graduate Teaching Assistant**, 6.886: Graph Analytics, MIT.
Spring 2016	**Undergraduate Teaching Assistant**, edX course on Big Data, MIT.

Outreach and Inclusion

2021, 2023	Volunteer at the Grace Hopper Celebration of Women in Computing.
2022	Mentor with the Berkeley AI Research Undergraduate Mentorship Program.
2019	Volunteer at the Google Summer of Code (formerly CodeU).